

IIT Foundation Module for Class VII Math

- Theme 1: Number System

1. Rational Numbers

- Properties of rational numbers (including identities). Using the general form of expression to describe properties
- Representation of rational numbers on the number line
- Between any two rational numbers, there lies another rational number
- Word problem

2. Exponents Powers

- Laws of exponents with integral powers
- Square and Square roots using the factor method and division method for numbers containing no more than 4 digits or 2 decimal places
- Cubes and cube roots (only factor method for numbers containing at most 3 digits)

3. Playing with numbers

- Writing and understanding 2 and 3-digit numbers in the generalised form ($100a + 10b + c$)
- Engaging with puzzles to solve and create problems
- Deducing divisibility test rules of 2, 3, 5, 9, 10 for 2 or 3-digit numbers

4. Sets

- Union and intersection of sets
- Disjoint-set
- Complement of a set

- Theme 2: Ratio and Proportion

1. Slightly advanced problems involving applications on percentages, profit & loss, overhead expenses, discount, tax
2. Difference between simple and compound interest (compounded yearly up to 3 years or half-yearly up to 3 steps only)
3. Direct and inverse variations – simple and direct word problems
4. Time and work problems – simple and direct word problems

- Theme 3: Algebra

- 1. Algebraic Expressions

- Multiplication and division of algebraic expressions (Coefficient should be integers)
 - Identities: $(a \pm b)^2 = a^2 \pm 2ab + b^2$, $a^2 - b^2 = (a - b)(a + b)$
 - Properties of inequalities
 - Factorisation (simple cases only): $a(x + y)$, $(x \pm y)^2$, $a^2 - b^2$, $(x + a)(x + b)$
 - Solving linear equations in one variable in contextual problems (avoid complex coefficients)

- Theme 4: Geometry

- 1. Understanding Shapes

- Properties of quadrilaterals – Angle Sum property
 - Properties of a parallelogram (By verification):
 - Opposite sides are equal
 - Opposite angles are equal
 - Diagonals bisect each other
 - Diagonals of a rectangle are equal and bisect each other
 - Diagonals of a rhombus bisect each other at right angles
 - Diagonals of a square are equal and bisect each other at right angles
 - Representing 3D in 2D
 - Identify and match pictures with objects (nested or joint 2D & 3D shapes, max 2)
 - Drawing 2D representation of 3D objects
 - Counting vertices, edges & faces & verifying Euler's relation for 3D figures (cubes, cuboids, tetrahedrons, prisms, pyramids)
 - Construction of Quadrilaterals
 - Given four sides and one diagonal
 - Three sides and two diagonals
 - Three sides and two included angles
 - Two adjacent sides and three angles
 - Reflection symmetry and symmetrical shapes
 - Circle: centre, radius/diameter, arc, chord, sector, and segment

- Theme 5: Mensuration

1. Area of a trapezium, a polygon, and semi-circle
2. Surface area of a cube, cuboid, cylinder
3. Total surface area and curved surface areas of various 3D figures
4. Concept of volume, measurement using basic units, volume of cube, cuboid, and cylinder
5. Volume and capacity (measurement of capacity)

- Theme 6: Data Handling

1. Arranging ungrouped data into groups
2. Representation of grouped data through bar-graphs, constructing and interpreting bar-graphs
3. Simple pie charts with reasonable data numbers
4. Consolidating and generalising the notion of chance in events like tossing coins, dice, etc., relating it to real-life events